

0-50V VARIABLE POWER SUPPLY

Using LM317

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Presented here is a 0-50V, 1A variable power supply that has been designed using positive-adjustable voltage regulator LM317 IC. This IC normally generates voltages in the range of 1.2V to 25V, but here it produces 0 to 50V with a current of 1A. The circuit used increases the upper voltage

from 25V to 50V and decreases the lower voltage from 1.2V to 0V.

Circuit and working

Fig. 1 shows the block diagram of the variable power supply. Mains input is connected to transformer X1. Output of the transformer is given to LM317 via rectifier diodes. Final output is provided by LM317 and transistor T1.

Heart of the circuit is LM317. Data sheet of LM317 IC states that it can produce voltages between 1.2V and 37V with a maximum current of 1.5A. LM317 has an internal reference voltage of 1.25V. This voltage is reflected in the output and is set

as the lower-limit voltage.

Maximum differential voltage between input and output should not exceed 40V to operate within safe limits. For this, voltage drop of 3V is maintained between input and output. To obtain maximum voltage with LM317, input should be 40V.

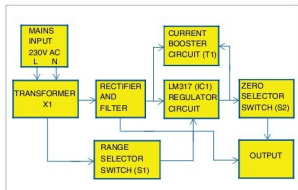


Fig. 1: Block diagram of the 0-50V, 1A variable power supply circuit

Test Points

Test point	Details
TP0	0V (GND)
TP1	1. Around 29V DC when S1 is at position 1 2. 56.4V DC when S1 is at position 2
TP2	1.2V-50V
TP3 when S2 is at 1.2V mode	1. 1.2V-27V, when S1 is at position 1 2. 27V-50V when S1 is at position 2
TP3 when S2 is at 0V mode	For measuring voltage below 4V

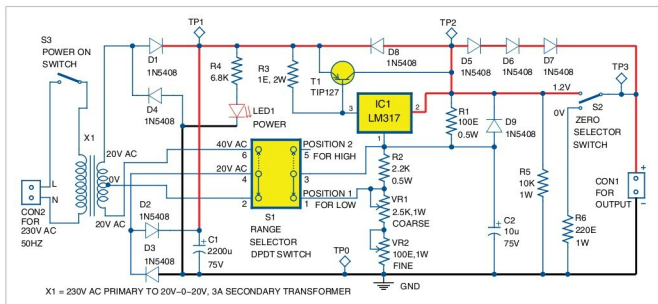


Fig. 2: Circuit diagram of the 0-50V, 1A variable power supply using LM317